

Smartphone Handwritten Circuits Solver Using Augmented Reality and Capsule Deep Networks for Engineering Education

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Introduction

- Circuit simulations are integral to engineering education and professional settings, but existing tools often require installation, access to workstations, and significant time, posing limitations for students.
- Machine learning, particularly capsule networks, offers a promising solution to make circuit learning easier by improving accuracy in recognizing hand-drawn circuits and by giving instant solutions.
- Capsule networks preserve spatial information between object parts, mimicking human recognition and effectively reducing the need for extensive training data.
- The paper proposes using capsule networks to design a mobile app for real-time analysis of hand-drawn circuits, aiming to reduce training data needs and enhance recognition accuracy.

Literature Review

Reference	Models	Performance	Limitations
Boraie, M.T.; A.S. “Optical recognition of electrical circuit drawings.”20–22 August 1997; Volume 2, pp. 843–846. [1]	Pixel-MI	Recognition accuracy: 80%	The Recognition accuracy is very low
Edwards, B.; Chandran, V. “Machine recognition of hand-drawn circuit diagrams.”5–9 June 2000; Volume 6, pp. 3618–3621. [2]	Pixel-MI	Component classification accuracy: 86%	Limitations in scaling to new components
Liu, Y.; Xiao, Y. Circuit Sketch Recognition; Department of Electrical Engineering Stanford University: Stanford, CA, USA, 2013. [3]	SVM-FD	83% recognition rate	Average accuracy
Rabbani, M.; Khoshkangini, R.; Nagendraswamy, H.; Conti, M. Hand Drawn Optical Circuit Recognition. Procedia Comput. Sci. 2016, 84, 41–48.[4]	ANN-Shape	Recognition of symbols, units, numbers, and alphabets	Low scalability to classify circuit elements
Sabour, S.; Frosst, N.; Hinton, G.E. Dynamic Routing Between Capsules. In Advances in Neural Information Processing Systems Curran Associates, Inc.: NewYork, NY, USA, 2017; pp. 3859–3869. [5]	CapsNet	Achieved good results on MNIST and Cifar10 datasets	10.6% error on Cifar-10 dataset due to background clutter

Contribution

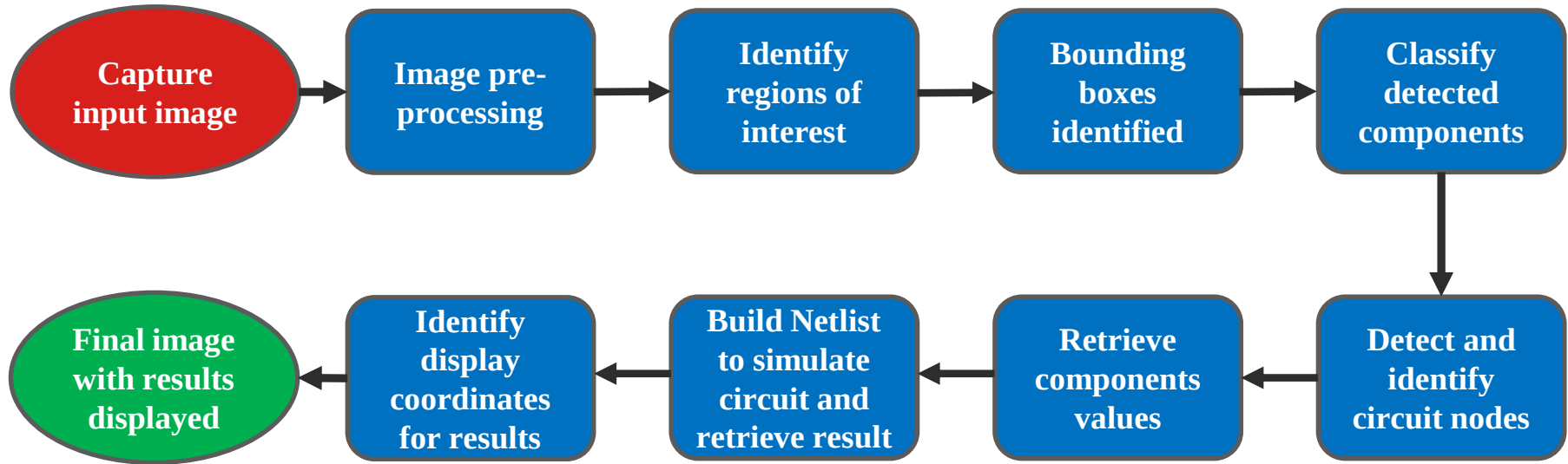
Key Contributions

Integration of machine learning techniques, particularly capsule networks, to recognize and detect circuit elements from images captured by smartphones.

Introduction of capsule networks as a novel technique in deep learning to significantly reduce the training data requirement for circuit recognition.

Development of a smartphone application for real-time analysis of hand-drawn circuits.

Methodology



Model Architecture

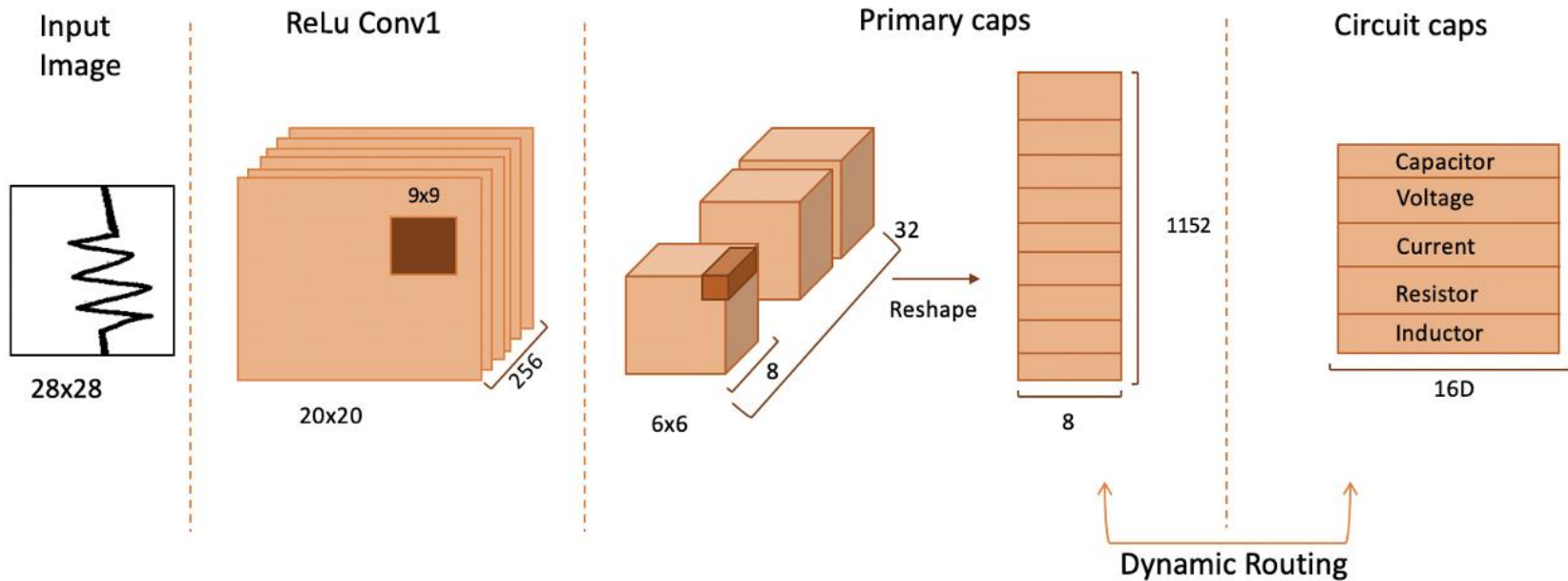
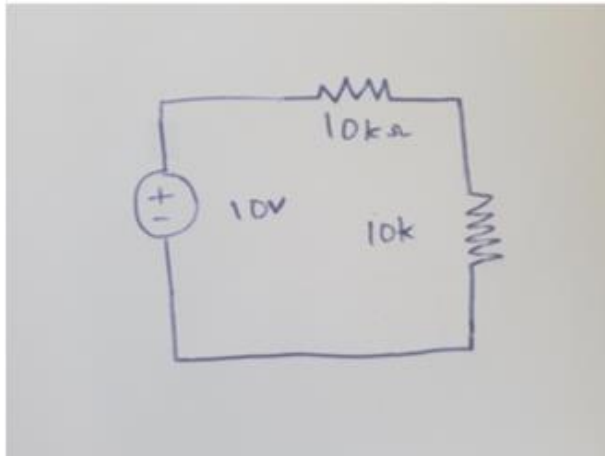
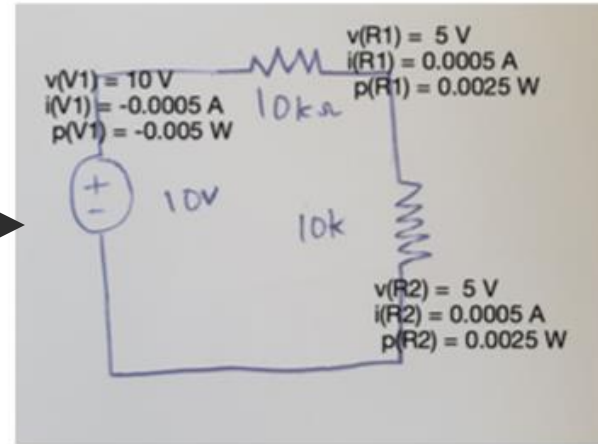


Fig. 1: Architecture of this proposed model.

Results



(a)



(b)

Fig. 2: (a) Test image (b) Overlay results on the test image

Performance analysis

	Precision	Recall	F1-Score
Capacitor	1.00	1.00	1.00
Current	0.92	0.91	0.91
Resistor	0.99	1.00	1.00
Voltage	0.92	0.89	0.90
Inductor	0.97	1.00	0.99
Micro Avg.	0.96	0.96	0.96
Macro Avg.	0.96	0.96	0.96
Weighted Avg.	0.96	0.96	0.96

Table 1: Performance analysis.

Future Work

- Refine object localization algorithm for varying circuit scales.
- Explore advanced techniques to enhance recognition accuracy.
- Improve usability and user experience of smartphone app.
- Gather feedback from educators and professionals for further enhancements.

Conclusion

- Introduce a smartphone app for direct digitization and get solution of hand-drawn circuit diagrams.
- Proposal of an augmented reality and capsule deep network solution for engineering education.
- Implementation of capsule networks due to their potential for high accuracy with less training data.
- Test results show CapsNet outperforms RCNN in object detection, indicating promise for circuit recognition.

References

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The background features a white space with abstract geometric elements. In the top right, there is a grey halftone pattern. A thick red line starts from the bottom left and trends upwards towards the top right, with several segments that are horizontal, creating a stepped or staircase effect. Several thin red lines are scattered across the slide, including one horizontal line in the top left and several diagonal lines in the bottom left and bottom right.

Thank you